Distill~~Coder~~Math: Capturing Large Model Performance in a Small Model via Distillation

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Why bother distilling?

Reasoning over memorization

Supervised Fine-Tuning teaches the model to memorize patterns by minimizing Cross-Entropy loss.

Distillation teaches the model to learn the reasoning of the teacher by adding a KL-Divergence term to the loss function.

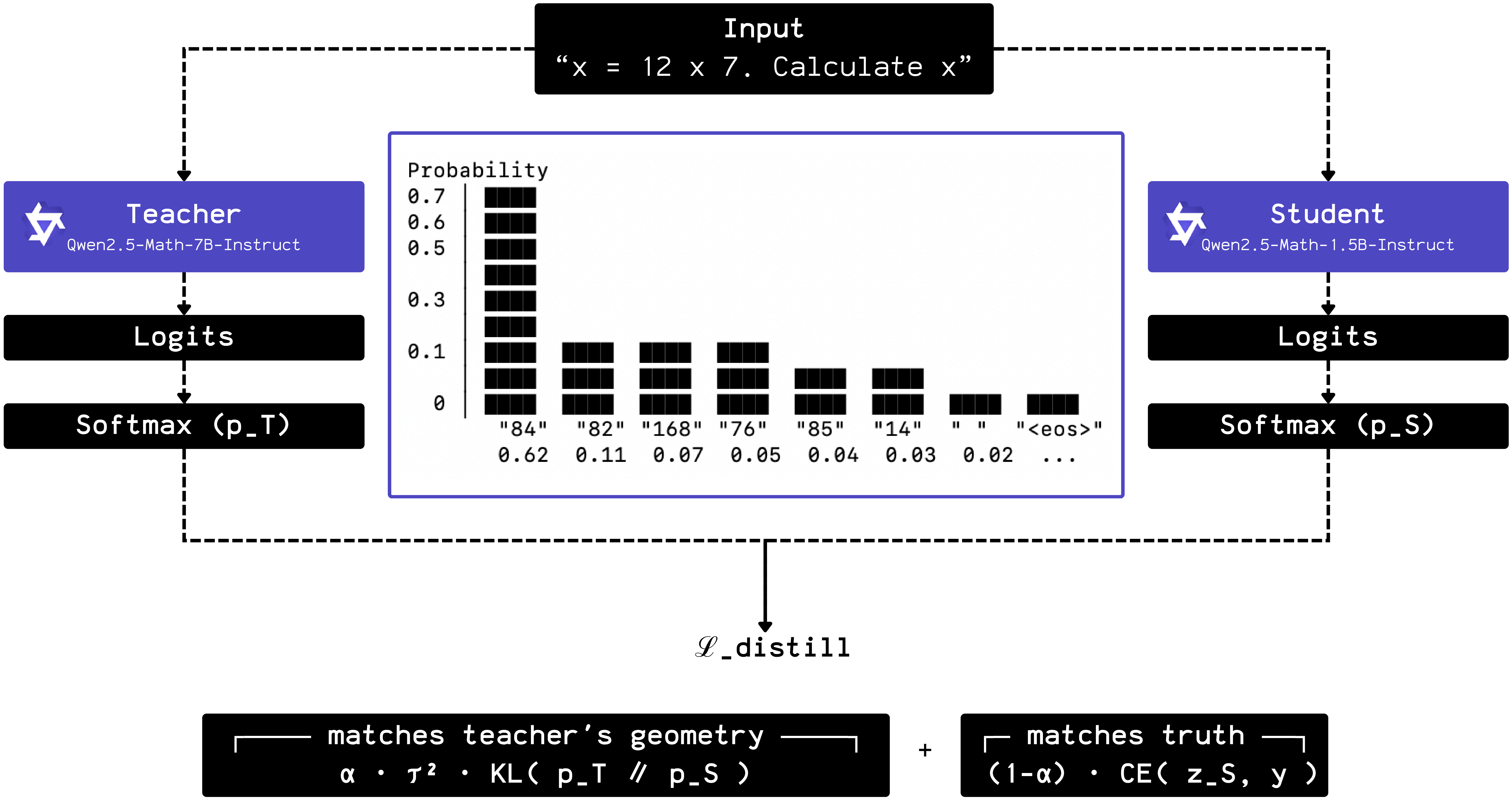
Where we started

Model	Performance on GSM8K	VRAM	Toks/s
Qwen2.5-Math-7B-Instruct	85.4%	14GB	~50
Qwen2.5-1.5B-Instruct	60-70%	3GB	~250

Table 4: **Performance of the 7B+ base models.**

Datasets	**Mistral-7B**	**Llama3-8B**	**Gemma2-9B**	**Qwen2-7B**	**Qwen2.5-7B**
Mathematics & Science Tasks					
GPQA	24.7	25.8	32.8	30.8	**36.4**
TheoremQA	19.2	22.1	28.9	29.6	**36.0**
MATH	10.2	20.5	37.7	43.5	**49.8**
MMLU-stem	50.1	55.3	65.1	64.2	**72.3**
GSM8K	36.2	55.3	70.7	80.2	**85.4**

The Process


Input
“x = 12 x 7. Calculate x”
Teacher
Qwen2.5-Math-7B-Instruct
Logits
Softmax (p_T)
Student
Qwen2.5-Math-1.5B-Instruct
Logits
Softmax (p_S)
Probability
0.7
0.6
0.5
0.3
0.1
0
"84" "82" "168" "76" "85" "14" " " "<eos>"
0.62 0.11 0.07 0.05 0.04 0.03 0.02 ...
ℒ_distill
matches teacher’s geometry
α · τ² · KL(p_T // p_S)
+
matches truth
(1-α) · CE(z_S, y)

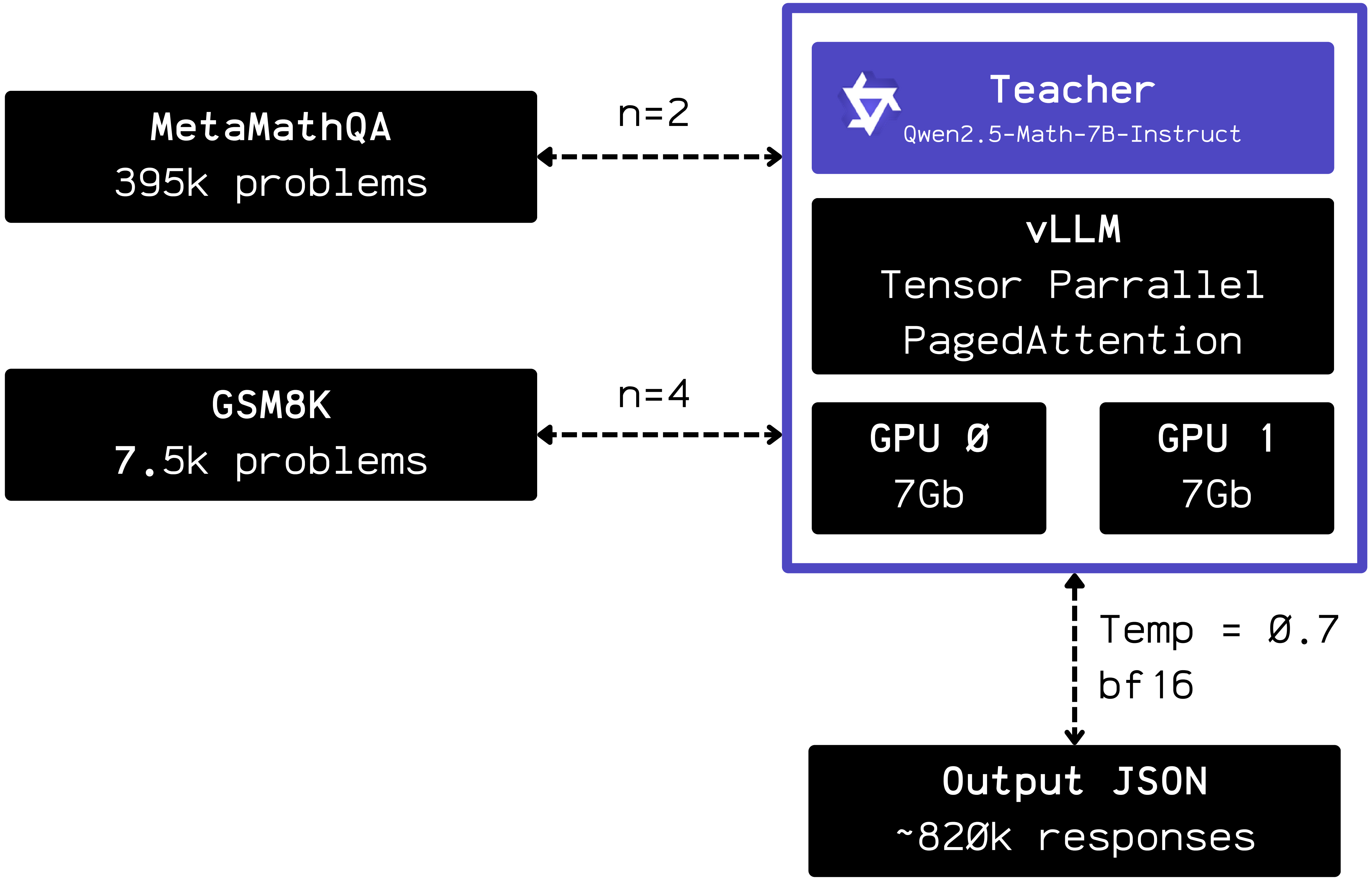
Pipeline

Collect
Teacher generates answers to math problems → JSON dataset

Distill
Student trains to match teacher's logits + ground-truth answer

Eval
Test student on GSM8K

Stage 1: Collection



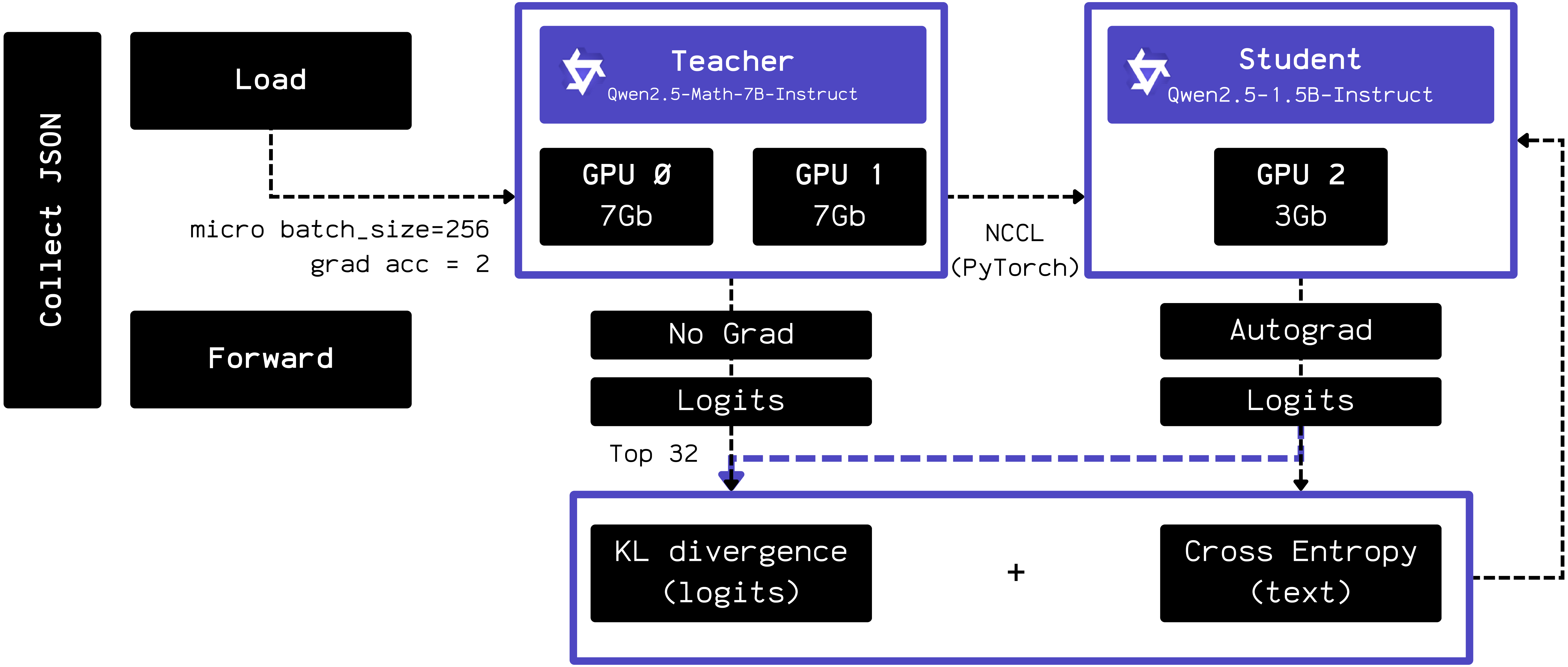
Prompt Template:

<|im_start|>system
Please reason step by step, and put your answer in \boxed{}.
<|im_end|><|im_start|>user
{QUESTION}<|im_end|>
<|im_start|>assistant

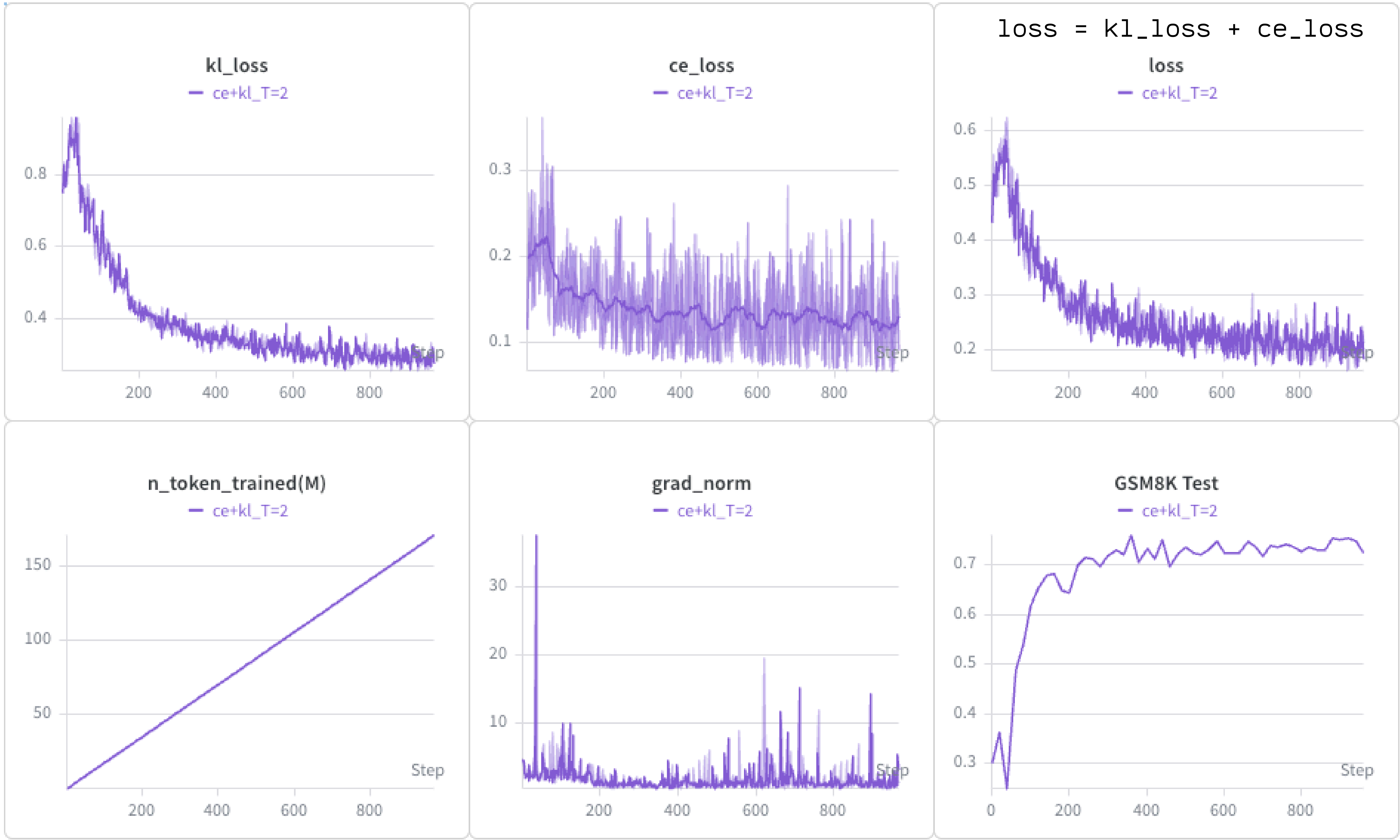
Output:

{prompt, [response_1, response_2, ...]}

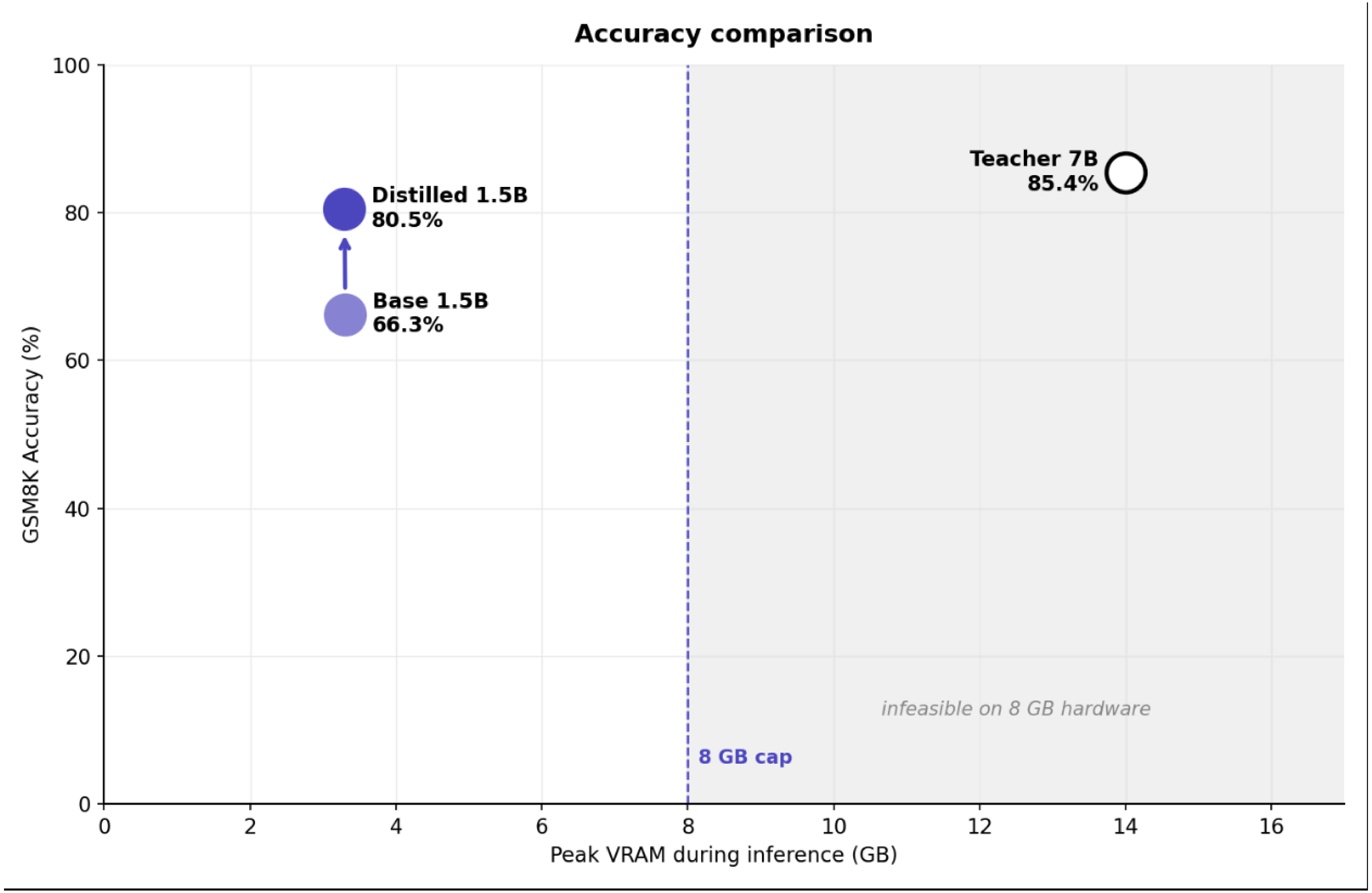
Stage 2: Distillation



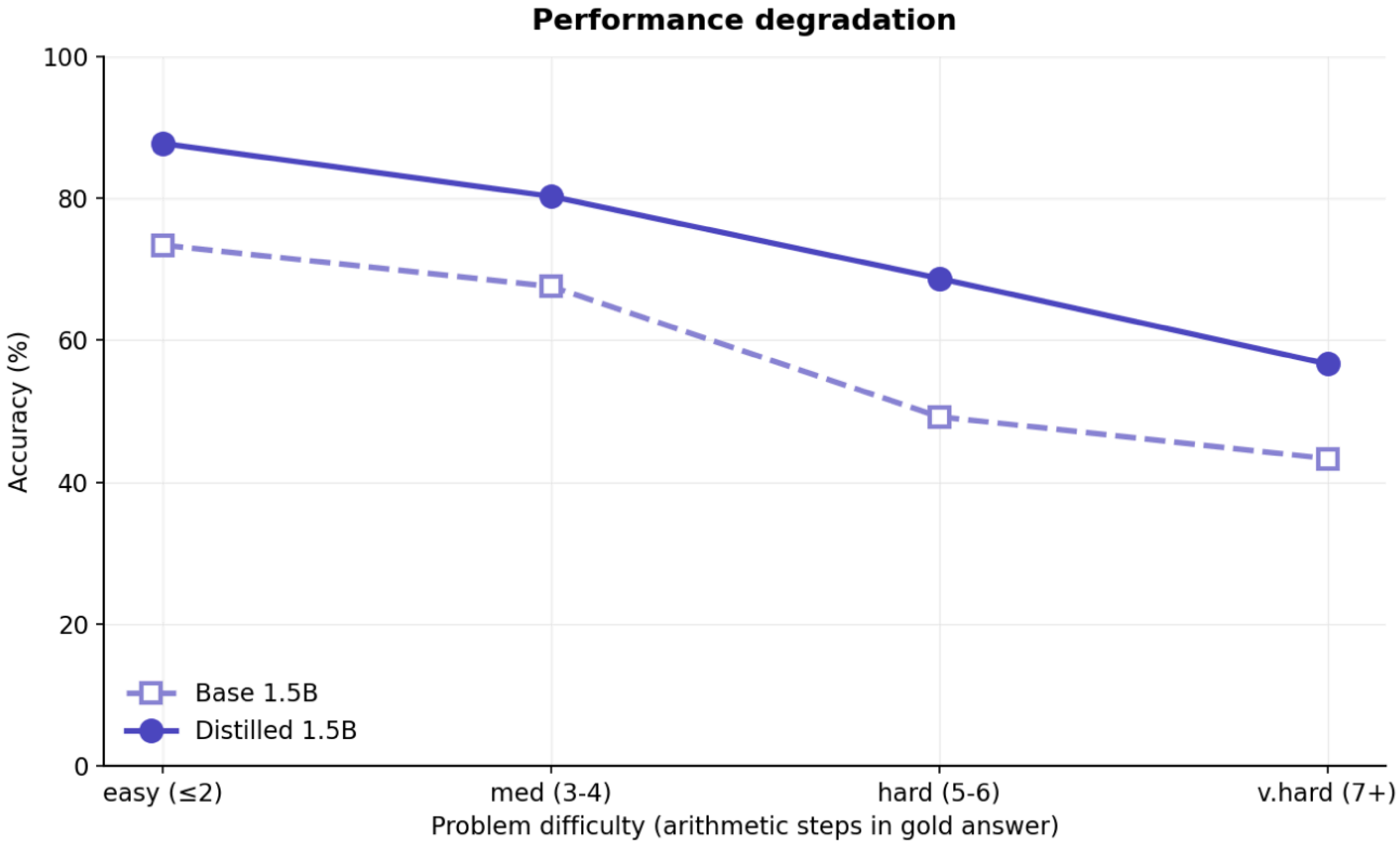
Training log



GSM8K - 1316 test examples - pass@1 - 8GB max VRAM



GSM8K - 1316 test examples - pass@1 - 8GB max VRAM



GSM8K - 1316 test examples - pass@1 - 8GB max VRAM

Metric	Teacher (Qwen-2.5-Math-7B-Instruct)	Student (Qwen-2.5-1.5B-Instruct)	Distilled (Ours)
Latency	-	16.9ms/tok	**17.1ms/tok**
Peak VRAM	OOM	3.31	**3.29GB**
Energy Avg (GPU) nvmlDeviceGetPowerUsage	-	114 W	**113 W**
Energy / correct	-	197 J	**158 J**
Accuracy	85.4%	66.3%	**80.5%**

"question": "A plague infects ten people. Every day, each infected person infects six others. How many people are infected after three days?",
"gold": "3430",
"responses": {
"Distilled 1.5B": {
"full": "To determine how many people are infected after three days, we need to follow the spread of the plague step by step.\n\n1. **Initial Condition:**\n - On day 0, there are 10 infected people.\n\n2. **Day 1:**\n - Each of the 10 infected people infects 6 others.\n - Number of new infections on day 1: \\(10 \\times 6 = 60\\)\n - Total number of infected people on day 1: \\(10 + 60 = 70\\)\n\n3. **Day 2:**\n - Each of the 70 infected people infects 6 others.\n - Number of new infections on day 2: \\(70 \\times 6 = 420\\)\n - Total number of infected people on day 2: \\(70 + 420 = 490\\)\n\n4. **Day 3:**\n - Each of the 490 infected people infects 6 others.\n - Number of new infections on day 3: \\(490 \\times 6 = 2940\\)\n - Total number of infected people on day 3: \\(490 + 2940 = 3430\\)\n\nThus, the total number of people infected after three days is \\(\\boxed{3430}\\).",
"predicted": "3430",
"correct": true
},
"Qwen 1.5B Instruct": {
"full": "To determine how many people are infected after three days, we need to follow the process of infection day by day.\n\n**Day 0:**\n- Initially, there is 1 infected person (the original plague).\n\n**Day 1:**\n- Each of the 1 infected person infects 6 new people.\n- Therefore, on Day 1, \\(1 \\times 6 = 6\\) new infections occur.\n- Total number of infected people after Day 1: \\(1 + 6 = 7\\).\n\n**Day 2:**\n- Each of the 7 infected people from Day 1 infects 6 new people.\n- Therefore, on Day 2, \\(7 \\times 6 = 42\\) new infections occur.\n- Total number of infected people after Day 2: \\(7 + 42 = 49\\).\n\n**Day 3:**\n- Each of the 49 infected people from Day 2 infects 6 new people.\n- Therefore, on Day 3, \\(49 \\times 6 = 294\\) new infections occur.\n- Total number of infected people after Day 3: \\(49 + 294 = 343\\).\n\nThus, the total number of people infected after three days is \\(\\boxed{343}\\).",
"predicted": "343",

ours

base

Demo(Video)

https://youtu.be/ks5diliFiXQ